

Related Worktable

Title	Authors	Years	Dataset	Application area	Method	Metrics	Limitations	Strengths	Research Gap	Relation to my work
Interpretable Deep Learning Framework for LULC Classification in Remote Sensing Using SHAP	Temenos et al.	2023	Interpretable Deep Learning Framework for LULC Classification in Remote Sensing Using SHAP	Land use/land cover classification with band-importance analysis	Compact CNN + SHAP (deep explainer)	Accuracy-94.72%	Accuracy capped below other CNN benchmarks (~98%); SHAP explanation is computationally expensive	Explainable band selection; reduces input data by 77% while keeping accuracy high; improves per-class accuracy	Most EuroSAT studies just report accuracy without explaining which spectral bands actually drive the predictions	Supports my project's use of explainability/interpretability alongside CNN classification on the EuroSAT dataset
Kolmogorov-Arnold Network for Satellite Image Classification in Remote Sensing	Cheon, M.	2024	EuroSAT - Sentinel-2 satellite images, 10 LULC classes	Satellite/land use image classification, exploring a novel MLP-alternative architecture	Kolmogorov-Arnold Network, tested alongside ConvNeXt, ResNet101, VGG16, MobileNetV2, EfficientNet, ViT	Best result: ConvNeXt + KAN reaching 96% Accuracy (starting at 94% in epoch 1)	Only a preprint accuracy (96%) is below top CNN/ViT benchmarks on EuroSAT (~98%+); KAN training is less mature/optimized than standard architectures	Introduces a newer, interpretable alternative to standard MLP classifier heads; benchmarks it against several established CNN/ViT backbones	KAN is still new and under-optimized for remote sensing, so its accuracy (96%) hasn't yet matched top CNN/ViT benchmarks (~98%+)	Shows me an alternative architecture I could reference or compare against if I explore beyond standard CNNs in my project

Deep Learning-Based Spatial Pattern Modeling for Land Use and Land Cover Classification Using Satellite Imagery	Marzougui, M. et al.	2025	EuroSAT - 27,000 Sentinel-2 images, 10 LULC classes	Spatial pattern modeling for LULC classification, environmental monitoring	Inception-V3 (fine-tuned hyperparameters)	Accuracy: 99.43%, Validation accuracy: 98.50%, Training loss: 0.0184, Validation loss: 0.0566	Single-architecture study (only tests Inception-V3, no comparison against other CNNs), published in a meteorology-focused journal rather than a CV/remote-sensing venue.	Very high accuracy (99.43%); reports both training and validation metrics, showing the model isn't overfitting	Only tests a single architecture (Inception-V3), so it doesn't compare performance against other CNN models on EuroSAT	Gives me a strong accuracy benchmark to compare my own model's performance against on the same EuroSAT dataset
Mapping of Land Use and Land Cover (LULC) using EuroSAT and Transfer Learning	Kunwar, S., Ferdush, J.	2023/2024	EuroSAT (RGB) - 27,000 images, 64×64 px, 10 classes	LULC mapping applied to a real region (Kreis Borken, Germany)	Vision Transformer (ViT) pre-trained on ImageNet-21k, fine-tuned via transfer learning.	Accuracy-99.19%	Uses only RGB bands transformer training needs GPU resources.	Highest accuracy in my related work (99.19%); demonstrates a real-world mapping use case beyond just classification	Only uses RGB bands and ignores the other 10 Sentinel-2 spectral bands, so multispectral information is left unused	Gives me a transfer-learning/transformer baseline to compare against if my project uses CNNs on the same RGB EuroSAT data

Benchmarking and Scaling of Deep Learning Models for Land Cover Image Classification	Papoutsis et al.	2023	BigEarthNet (Sentinel-2, multilabel) — evaluated further on EuroSAT and SEN12MS	Large-scale LULC benchmarking and transfer learning	CNN zoo (60+ models), EfficientNet-scaled Wide ResNet + Efficient Channel Attention	+4.5% F1score over ResNet50 baseline (BigEarthNet, 19 classes)	EuroSAT is a secondary evaluation, not primary focus; largescale setup is compute-heavy	Extensive model zoo (60+ models); domain-relevant pretrained weights improve transfer performance to EuroSAT	EuroSAT is only a secondary evaluation dataset here, so there isn't deep EuroSAT-specific analysis	Suggests pretraining on a larger remote-sensing dataset before fine-tuning could improve my model's accuracy on EuroSAT
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